

scripts/autonomous-qa/run-iteration.sh — User Guide

Last verified: 2026-07-26 (LVA-018 script-docs backfill cycle)

Inheritance: HelixConstitution §11.4.18 (script docs); Lava §6.J / §6.L (anti-bluff), §6.AK (device-gate auth), §6.H (credential handling)

Classification: project-specific

Overview

Runs **one QA iteration** against an already-booted emulator + already-up backend (the matrix cell executed by `run-matrix.sh`):

1. **FRESH client install** (`uninstall + install -r`) — clean onboarding state
2. Start recording: cleared logcat + chunked screenrecord (180 s cap/chunk)
3. Run `Challenge70AutonomousQaProviderMatrixTest` parameterized via gradle `connectedDebugAndroidTest` instrumentation arguments
4. Stop recording (+ a best-effort `final-screen.png` still while the app is up)
5. Parse the JUnit verdict → `curated verdict.json`

Raw media/logs go to `<evidence-dir>/raw/` (gitignored); `verdict.json` + `junit.xml` are curated.

Usage

```
scripts/autonomous-qa/run-iteration.sh \  
  --backend goapi|apiapp --providers <csv> --query <str> \  
  --serial <adb-serial> --api-url <url> --evidence-dir <dir>
```

All of `--backend`, `--providers`, `--query`, `--serial`, `--evidence-dir` are required; `--api-url` defaults to `https://127.0.0.1:8443`.

The goapi auth key (§6.AK / §6.H)

For the goapi backend the client must authenticate /v1 with the Go API's configured Lava-Auth key. The script derives it from the same root .env the compose stack feeds the container (LAVA_AUTH_ACTIVE_CLIENTS: name:uuid) — it needs the UUID only, NOT the HMAC secret. The derived key is credential-equivalent and is **never echoed**. Without an entry the iteration runs keyless (and /v1 will 401 — recorded honestly).

Verdict decision table (anti-bluff)

The verdict is decided by the gradle/JUnit outcome **alone**. A crash is a failure: no signature, marker, or "known defect" converts a failed run into a PASS.

Condition	Verdict
gradle_rc=0, 0 failures/errors, tests>0, not all skipped	PASS
gradle_rc=0 but tests=0	FAIL — a green run with zero executed tests is a bluff by construction
all tests skipped	SKIP
anything else (including any crash)	FAIL

Two signals are grepped from logcat + JUnit XML and recorded as **diagnostic evidence only**. They describe what happened during the run; they do **not** gate the verdict:

- marker_download_ok — Challenge70 logged C70-RESULT ... DOWNLOAD-OK, i.e. the on-screen download/magnet affordance was confirmed
- other_failure_signal — an explicit C70 download-step failure marker, or any AssertionError. No phrase is whitelisted.

Removed: the LVA-008 PASS-override (do not re-introduce)

A teardown_known_lva008 signal and a PASS-via-marker branch used to convert a run into a PASS when the raw stream carried both LVA-008 teardown phrases (State must be at least 'CREATED' + Unable to

destroy activity), on the premise that the crash was an unfixable-upstream AndroidX defect. That premise was **disproven**: LVA-008 was a Lava threading bug (an off-main-thread `navigate()` call) fixed on 2026-06-30 by commit `ccdd84c1`. With the cause fixed, the override was a live mechanism for reporting a genuine crash as green — the canonical §6.J bluff — and was removed on 2026-08-26. If that signature appears again it is a **regression** and must fail the iteration.

A missing client APK writes a `FAIL verdict.json` immediately (never leaves a stale `PASS` behind — the 2026-07-03 incident).

Exit codes

- 0 — verdict `PASS`
- 1 — verdict `FAIL/SKIP`
- 2 — usage error or setup error (e.g. missing APK; a `FAIL` verdict is written)

`verdict.json` fields

`backend`, `providers`, `query`, `serial`, `gradle_rc`, `tests`, `failures`, `errors`, `skipped`, `marker_download_ok`, `other_failure_signal`, `verdict`, `note`, `junit_xml`, `raw_dir`.

Companion files

- `scripts/autonomous-qa/run-matrix.sh` — the caller
- `scripts/autonomous-qa/lib-summary.sh` — parses these verdict files back out
- `scripts/autonomous-qa/vision-analyze.sh` — analyzes the raw/recordings